Multimodal Brain Imaging Using a Hybrid Deep Learning Framework for CT and MRI Fusion

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***Abstract*—Multimodal medical image fusion is an effective technique that improves diagnostic accuracy by integrating complementary information from Computed Tomography (CT) and Magnetic Resonance Imaging (MRI). This paper introduces a hybrid deep learning–based fusion framework that combines structured feature extraction with adaptive fusion mechanisms to optimize the preservation of detail and overall information content in the fused images. Unlike conventional CNN-based and GAN-based methods, the proposed approach emphasizes structural correspondence and balanced intensity distribution, resulting in enhanced visualization and more clinically relevant features. The model is evaluated using quantitative metrics, including Entropy, Mutual Information (MI), Structural Similarity Index (SSIM), Peak Signal-to-Noise Ratio (PSNR), and Root Mean Square Error (RMSE). Experimental results demonstrate that the proposed method achieves higher Entropy, MI, SSIM, and PSNR values, along with lower RMSE levels, compared to existing CNN- and GAN-based fusion techniques. These findings indicate superior feature preservation, reduced distortion, and improved overall image quality across a variety of CT–MRI samples, highlighting its potential for practical clinical applications.**

***Index Terms*—Medical image fusion, CT-MRI fusion, hybrid CNN-Transformer, feature-level fusion, unsupervised learning, diagnostic imaging, structural preservation**

I. Introduction

Medical imaging plays a critical role in clinical diagnosis, treatment planning, and disease monitoring [1,2]. Different imaging modalities provide complementary information due to variations in physical acquisition principles and tissue sensitivity. Computed Tomography (CT) produces high-resolution structural representations, particularly effective for visualizing dense tissues such as bones and calcifications [1,2]. In contrast, Magnetic Resonance Imaging (MRI) provides superior soft-tissue contrast and enhanced visualization of organs and pathological regions [1]. Relying on a single modality may therefore lead to incomplete diagnostic interpretation.

Multimodal medical image fusion aims to integrate complementary information from multiple imaging sources into a single composite image that simultaneously preserves structural details and soft-tissue contrast [3]. An effective fusion technique should retain the anatomical sharpness of CT images while maintaining the intensity discrimination of MRI, thereby improving diagnostic reliability and reducing cognitive burden on clinicians.

Traditional image fusion approaches, including multi-scale transform and pyramid-based methods, rely on handcrafted feature extraction and predefined fusion rules. Although computationally simple, these techniques often fail to maintain a balanced representation of modality-specific characteristics. Recent progress in deep neural architectures has enabled Convolutional Neural Networks (CNNs) to effectively learn hierarchical spatial representations for image fusion tasks [4]. However, CNNs operate primarily within local receptive fields and may not effectively model long-range contextual dependencies across the entire image.

Generative Adversarial Network (GAN)-based fusion methods have also been proposed to enhance perceptual quality through adversarial learning [5]. While GANs improve visual realism, they often suffer from training instability and may introduce artificial artifacts. Recently, Transformer architectures have shown strong capability in modeling global relationships using self-attention mechanisms [6]. Integrating Transformer-based contextual modeling with CNN-based local feature extraction offers a promising approach to overcome the limitations of existing fusion methods [3].

Motivated by these observations, this work proposes a hybrid CNN–Transformer framework for CT–MRI image fusion. The convolutional layers capture fine-grained spatial structures, while the Transformer encoder models long-range dependencies to ensure global contextual consistency. A pseudo-supervised training strategy based on pixel-wise intensity averaging is adopted to eliminate the need for ground-truth fused images [11]. The proposed approach is evaluated against CNN-based, GAN-based, and direct pixel-averaging fusion methods

using multiple quantitative metrics to validate its effectiveness.

A. Main Contributions

This work makes the following technical contributions:

- A CNN-Transformer based hybrid architecture for CT-MRI image fusion which combines local spatial feature extraction and long-range dependency modeling.
- A feature-level fusion strategy that ensures balanced modality contribution while maintaining structural integrity and intensity consistency.
- A pseudo-supervised training mechanism based on pixel-wise intensity averaging, eliminating the requirement for ground-truth fused images.
- A comprehensive comparative analysis against CNN-based and GAN-based fusion methods, along with direct pixel-wise averaging, using multiple quantitative metrics including Entropy, Mutual Information (MI), SSIM, PSNR, and RMSE.

II. Related Work

Recent advances in medical image fusion have been driven primarily by deep learning frameworks [1]. Existing approaches can be broadly categorized into CNN-based methods, GAN-based models [2], Transformer-based architectures [3], and diffusion-driven fusion strategies [4].

CNN-based fusion frameworks have demonstrated strong capability in hierarchical feature extraction and structural preservation [5]. Architectures incorporating residual connections, multi-scale feature extraction, and entropy-guided mechanisms improve edge clarity and anatomical boundary retention. However, CNNs operate within limited receptive fields and may struggle to capture long-range contextual dependencies effectively [6].

GAN-based fusion models introduce adversarial learning to enhance perceptual realism and structural consistency. Frameworks such as structured constraint GANs and unsupervised medical fusion GANs demonstrate improved PSNR and SSIM performance. Despite these improvements, GAN-based methods often suffer from training instability, mode collapse, and potential artifact generation, which may limit their reliability in clinical deployment [2,7].

In recent years transformer-based and hybrid architectures have received significant scholarly attention due to their ability to model global contextual interactions using self-attention mechanisms. Parallel transformer structures and attention-guided fusion networks effectively preserve global-local feature relationships. However, increased computational complexity and optimization sensitivity remain significant challenges [8].

Diffusion-based fusion models further enhance structural preservation through progressive denoising and multiscale attention mechanisms. While these approaches achieve state-of-the-art quantitative performance, their iterative refinement process increases computational overhead [9].

Although these methods demonstrate promising results, achieving stable multimodal fusion that simultaneously preserves local anatomical details and global contextual consistency without requiring ground-truth fused images remains challenging [6,7]. To address these limitations, a hybrid CNN–Transformer framework is developed and trained using a pseudo-supervised strategy to jointly preserve structural integrity, enhance contextual representation, and maintain computational efficiency.

III. Proposed Methodology

The proposed hybrid CNN–Transformer framework is designed to integrate complementary structural and contextual information from CT and MRI modalities. The architecture consists of four main stages: preprocessing, feature extraction, feature-level fusion, and reconstruction.

A. Problem Formulation

Let the CT image be represented as:

$$I_{CT} \in \mathbb{R}^{1\times H\times W} \tag{1}$$

and the MRI image as:

$$I_{MRI} \in \mathbb{R}^{1\times H\times W} \tag{2}$$

where W and H denote the spatial dimensions of the image in terms of width and height, respectively.

Multimodal fusion seeks to establish a functional relationship between input modalities, expressed as:

$$F : (I_{CT}, I_{MRI}) \rightarrow I_{fused} \tag{3}$$

such that the fused image I_{fused} preserves anatomical structures from CT and soft-tissue contrast from MRI.

B. Network Architecture

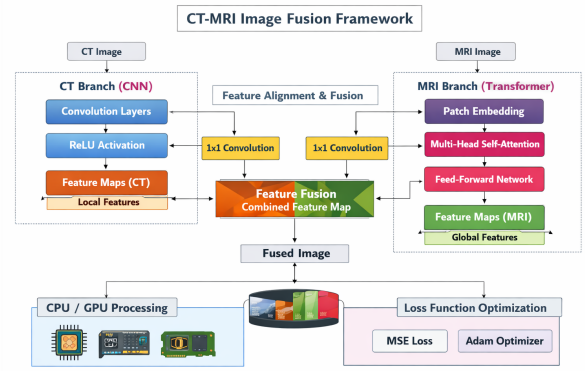


Fig. 1. Proposed hybrid CNN–Transformer architecture

Figure 1 depicts the overall design of the proposed framework, which is organized into two parallel processing streams to handle modality-specific representation learning. The CT

stream employs convolutional operations to capture fine-grained spatial structures, whereas the MRI stream incorporates a Transformer encoder to model long-range contextual interactions across the image.

C. CNN-Based Local Feature Extraction

For the CT input, spatial representations are obtained through two successive convolutional layers configured with a 3 × 3 kernel and unit stride, each followed by a ReLU activation function for nonlinearity:

$$F_{CT}^{(1)} = ReLU(Conv_{3\times3}(I_{CT})) \tag{4}$$

$$F_{CT} = ReLU(Conv_{3\times3}(F_{CT}^{(1)})) \tag{5}$$

The first convolution layer expands the channel dimension from 1 to 16, and the second increases it to 32 channels. This branch captures fine-grained spatial details such as edges, textures, and anatomical boundaries.

D. Transformer-Based Global Feature Modeling

In the MRI processing branch, the input image is first projected into an enhanced feature representation using a 1×1 convolutional layer:

$$X = Conv_{1\times1}(I_{MRI}) \tag{6}$$

The resulting feature map X is then flattened into a token sequence and processed by a Transformer encoder incorporating multi-head self-attention. The self-attention operation can be expressed as:

$$Attention(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{7}$$

where Q, K, and V represent the query, key, and value projections, respectively, and d_k denotes the dimensional scaling factor.

Through this attention-driven mechanism, the encoder captures long-range spatial interactions and global contextual relationships. The refined token embeddings are subsequently rearranged into spatial form to produce the contextual feature representation F_{MRI}.

E. Feature-Level Fusion

To integrate complementary information from both modalities, feature-level averaging is performed:

$$F_{fused} = \frac{F_{CT} + F_{MRI}}{2} \tag{8}$$

To ensure dimensional consistency between feature maps, a 1 × 1 convolution is utilized to reconcile channel depth prior to integration. This fusion strategy ensures balanced modality contribution while maintaining structural consistency.

F. Reconstruction

The fused feature representation is projected back to image space using a convolutional reconstruction layer:

$$I_{fused} = Conv_{3\times3}(F_{fused}) \tag{9}$$

This step generates the final fused image while preserving both local structural details and global contextual information.

IV. Experimental Setup

A. Dataset and Preprocessing

The Harvard Medical Image Fusion dataset available on Kaggle is used for experimental evaluation. The dataset consists of 184 paired CT and MRI brain images acquired under clinically validated conditions.For experimental validation, 80% of the dataset (147 image pairs) is utilized for training, whereas the remaining 20% (37 image pairs) is used for testing. Before training, each image undergoes grayscale transformation followed by intensity scaling to the interval $[0, 1]$ to improve optimization stability. To ensure uniform input dimensions, all samples are resampled to a predefined spatial resolution. The processed images are then converted into PyTorch tensors of size $(B, 1, H, W)$, where B denotes the batch size, enabling efficient batch-wise processing.

B. Training Strategy

Since ground-truth fused images are not available in medical fusion datasets, a pseudo-supervised learning strategy is adopted. The reference target is defined as the pixel-wise intensity average of the two input modalities:

$$I_{target} = \frac{I_{CT} + I_{MRI}}{2} \tag{10}$$

The optimization objective is defined using Mean Squared Error (MSE) loss:

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}\left(I_{fused}^{(i)} - I_{target}^{(i)}\right)^2 \tag{11}$$

where N denotes the total number of pixels in the image. This formulation ensures intensity consistency while allowing the network to learn structural integration through feature extraction mechanisms.

The model is trained using the Adam optimizer with an initial learning rate of 1×10^{-3}. Training is performed for 20 epochs with mini-batch optimization. No adversarial or auxiliary losses are used, ensuring stable convergence during training.

V. Results and Discussion

The proposed hybrid CNN–Transformer model was evaluated using ten paired CT and MRI images. The performance of the proposed method was compared with two commonly used image fusion approaches, namely a CNN-based model and a GAN-based model. Both qualitative and quantitative analyses were conducted to evaluate fusion effectiveness.

A. Qualitative Results and Visual Analysis

1. CT and MRI Input Images

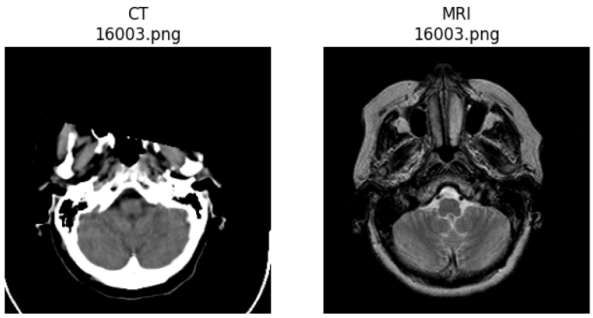
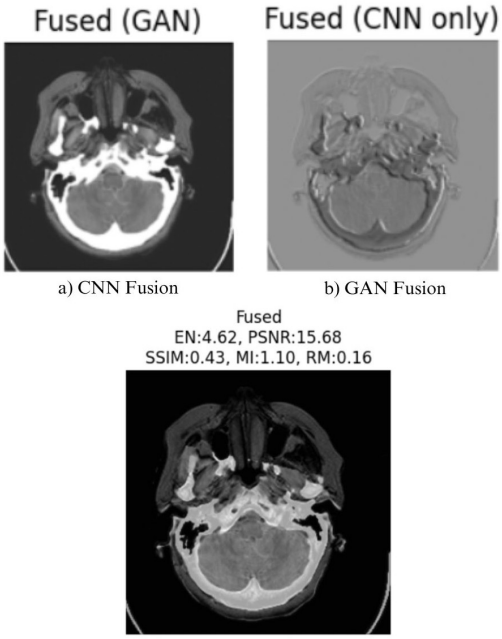


Fig. 2. CT and MRI input images

2. Comparison of Existing Models and Proposed Models



a) CNN Fusion b) GAN Fusion

c) Proposed Model

Fig. 3. Fusion results comparison: (a) CNN, (b) GAN, (c) Hybrid CNN–Transformer.

The superiority of the proposed methodology over CNN-only and GAN-based fusion approaches is clearly demonstrated in Fig. 3. While the GAN-based result appears sharper, it introduces intensity imbalance and artifacts in certain regions. The CNN-based fused image exhibits reduced contrast and incomplete structural representation.

In contrast, the proposed hybrid CNN–Transformer approach preserves both soft-tissue and bone details while enhancing structural clarity and maintaining balanced intensity distribution. The higher Entropy and PSNR values, along with improved SSIM and MI metrics, further validate this qualitative improvement, indicating superior visual quality and better information retention compared to the CNN and GAN models.

B. Quantitative Analysis

The performance metrics used in this study include Entropy, Mutual Information (MI), Structural Similarity Index Measure (SSIM), Peak Signal-to-Noise Ratio (PSNR), and Root Mean Square Error (RMSE). These metrics collectively measure information preservation, structural consistency, reconstruction quality, and error minimization.

TABLE I
QUANTITATIVE PERFORMANCE COMPARISON OF IMAGE FUSION METHODS

METRICS	CNN	GAN	Proposed Model
Entropy	1.081	4.177	4.482
PSNR (dB)	8.66	7.43	15.34
SSIM	0.125	0.289	0.528
MI	0.0187	1.089	1.141
RMSE	0.369	0.427	0.172

From Table I, it can be observed that the proposed model attains the highest Entropy (4.482), indicating superior information retention. The PSNR value (15.34) significantly exceeds that of CNN and GAN, demonstrating improved reconstruction quality. The SSIM value (0.528) confirms better structural similarity between the fused and source images. The Mutual Information (1.141) is higher than both CNN and GAN, reflecting stronger integration of complementary modality information. Furthermore, the proposed model records the lowest RMSE (0.172), indicating reduced reconstruction error and enhanced fusion accuracy.

C. Graphical Comparison

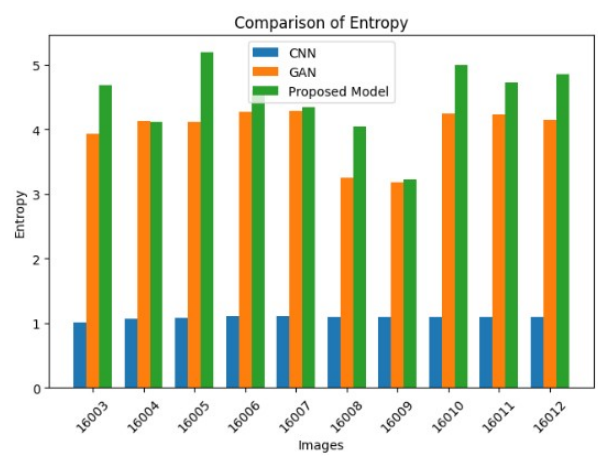


Fig. 4. Entropy visualization analysis.

Entropy (EN): Figure 4 illustrates the entropy values of the fused CT–MRI images in the test samples. Entropy measures the amount of information preserved in the fused image. The proposed hybrid CNN–Transformer model achieves the highest average entropy (4.482), outperforming the CNN-based (1.081) and GAN-based (4.177) fusion methods. This indicates the hybrid model better preserves modality-specific information and structural details.

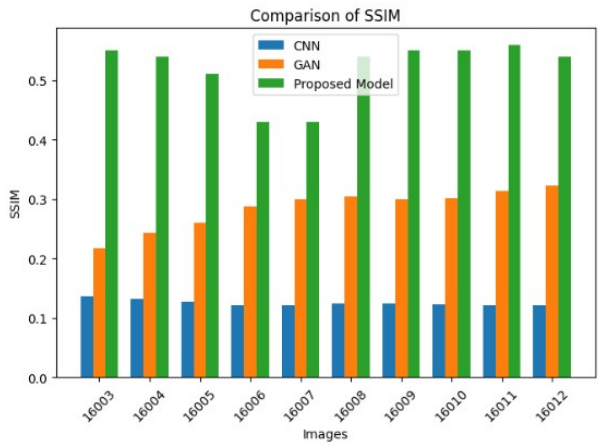


Fig. 5. SSIM visualization analysis.

Structural Similarity Index (SSIM): Figure 5 presents the SSIM values for the same test images. SSIM evaluates how well the structural information from the source images is preserved in the fused output. The hybrid CNN–Transformer model consistently achieves the highest average SSIM (0.53, range 0.43–0.56), compared to CNN (0.13) and GAN (0.29). This shows improved structural preservation in soft tissue areas.

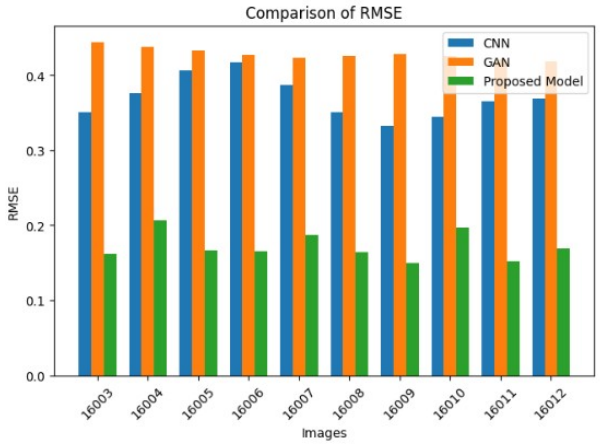


Fig. 6. RMSE visualization analysis.

Root Mean Square Error (RMSE): Figure 6 shows the RMSE values across the test images. RMSE quantifies the pixel-wise reconstruction error in the fused image. The proposed model achieves the lowest average RMSE (0.172, range 0.15–0.21), significantly better than CNN (0.37) and GAN (0.43). This confirms the hybrid model produces high-fidelity fused images with reduced distortion.

D. Discussion

Combining CNN-based local feature extraction and Transformer-based global attention for enhancing CT-MRI fusion. The hybrid model outperforms standalone CNN and GAN approaches in structural preservation, information retention, and reconstruction accuracy. Future work includes testing on larger datasets and optimizing for real-time deployment.

VI. Conclusion

This paper presented a hybrid CNN–Transformer framework for CT–MRI image fusion that integrates local spatial feature extraction with global contextual modeling. The proposed feature-level averaging strategy enables balanced modality contribution while preserving structural integrity and intensity consistency. The experimental findings prove that the proposed method is more effective than CNN-based and GAN-based types of fusion in terms of Entropy, PSNR, SSIM, mutual information and RMSE. The hybrid architecture effectively enhances structural preservation, information retention, and reconstruction accuracy.

VII. Future Work

Future studies will assess the hybrid CNN Transformer framework on bigger and more varied clinical data, optimize it to be used in real time, consider advanced attention and multi-scale feature fusion methods, and examine the explainable AI methods to increase interpretability and clinical confidence.

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